

System Architecture Specification

Problem Statement: E-Commerce Platform Order Management

UML Diagram 01 / 04

1. USE CASE DIAGRAM

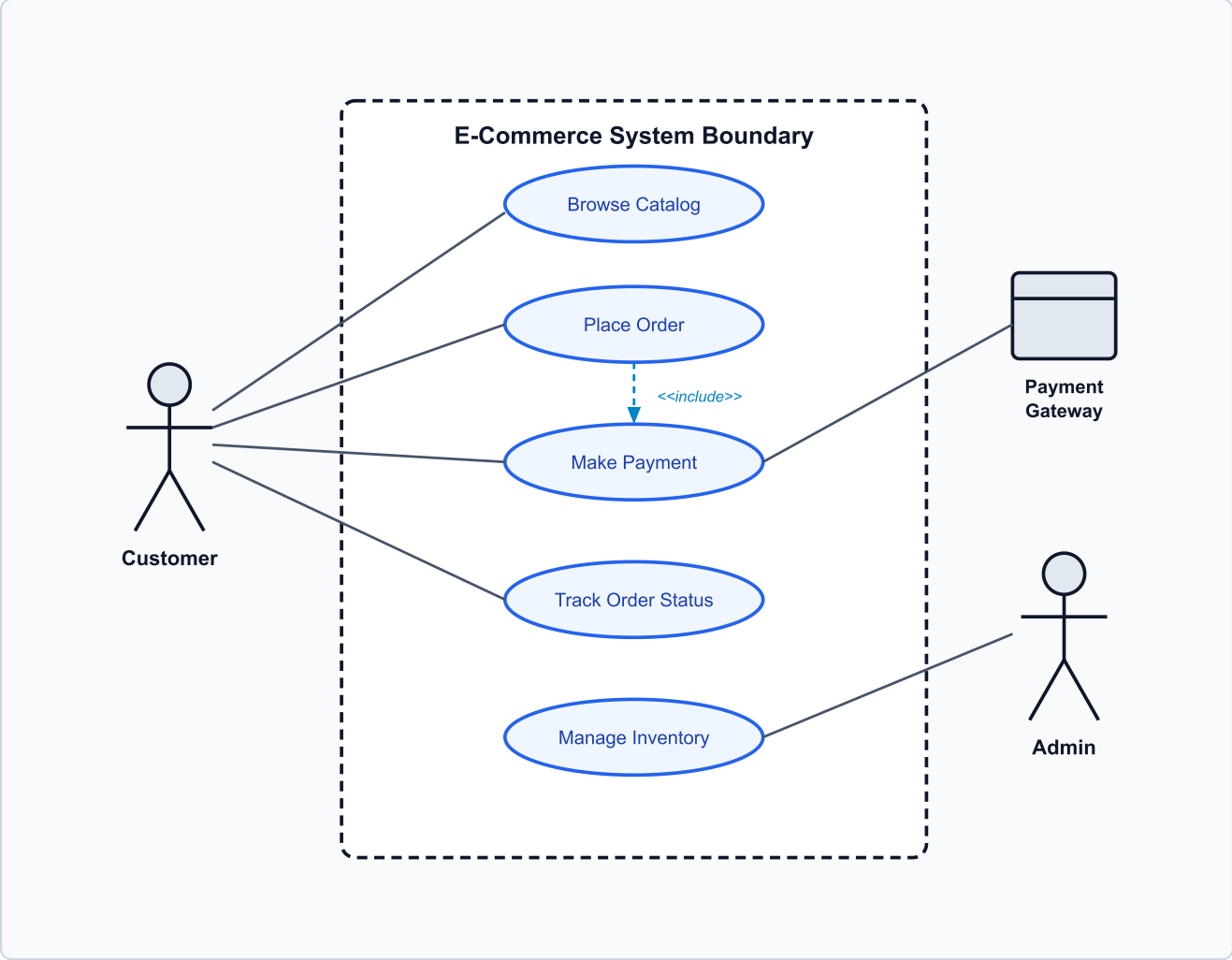


Diagram Key Highlights:

- Primary Actors:** Customer (initiates catalog browsing, order placement, and tracking) and Administrator (manages catalog and inventory).
- Secondary Actor:** External Payment Gateway API to process card and digital payment transactions.
- Relationships:** Placing an order enforces a mandatory `<<include>>` dependency on the Payment processing use case.

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UML Diagram 02 / 04

2. CLASS DIAGRAM

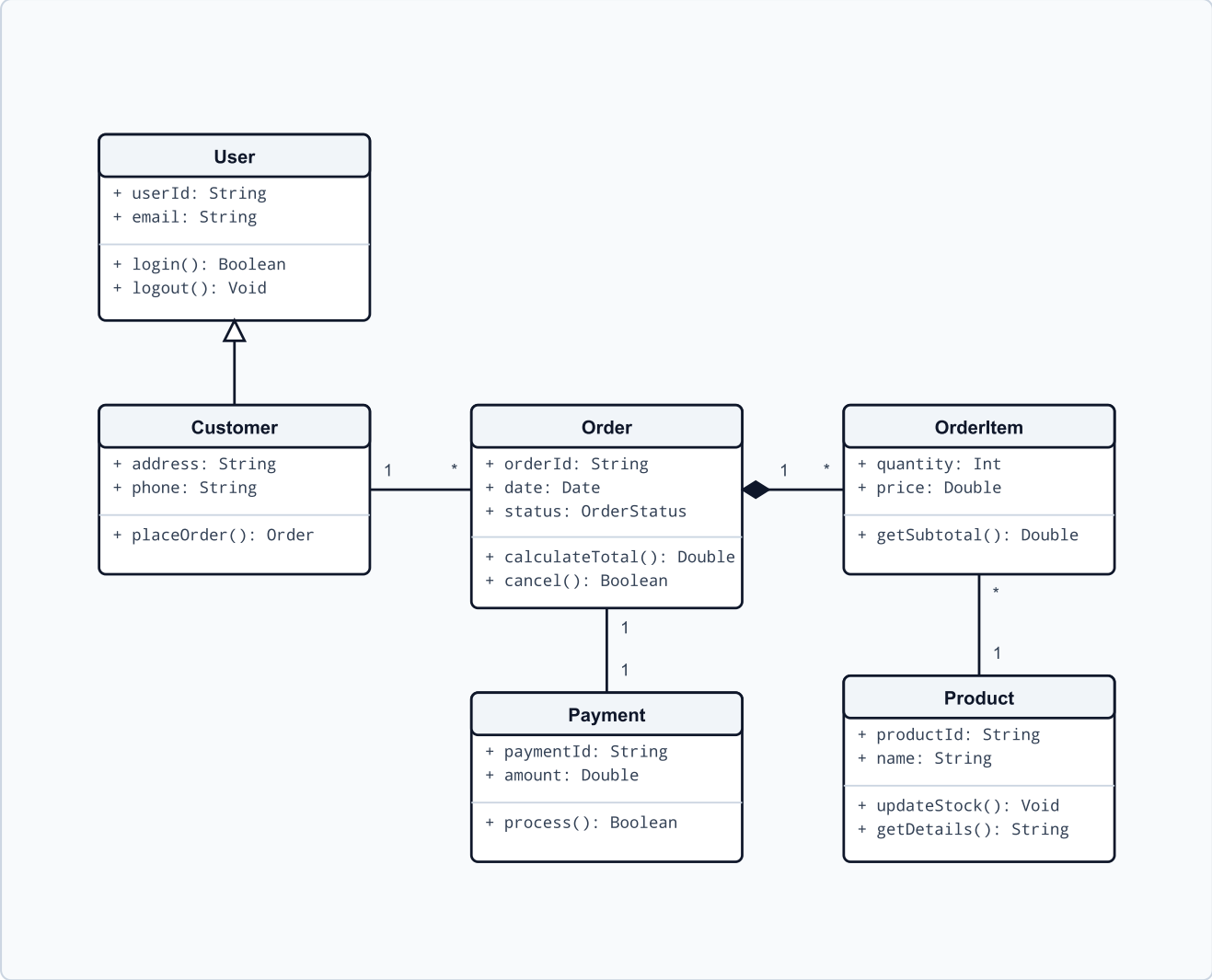


Diagram Key Highlights:

- Inheritance:** Customer inherits general credentials and methods from base class User.
- Composition:** Order contains one or more OrderItem objects; line items cannot exist independently of an Order.
- Associations:** A Customer can place multiple Orders (1 to *); an Order maps directly to 1 Payment instance.

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UML Diagram 03 / 04

3. COMPONENT DIAGRAM

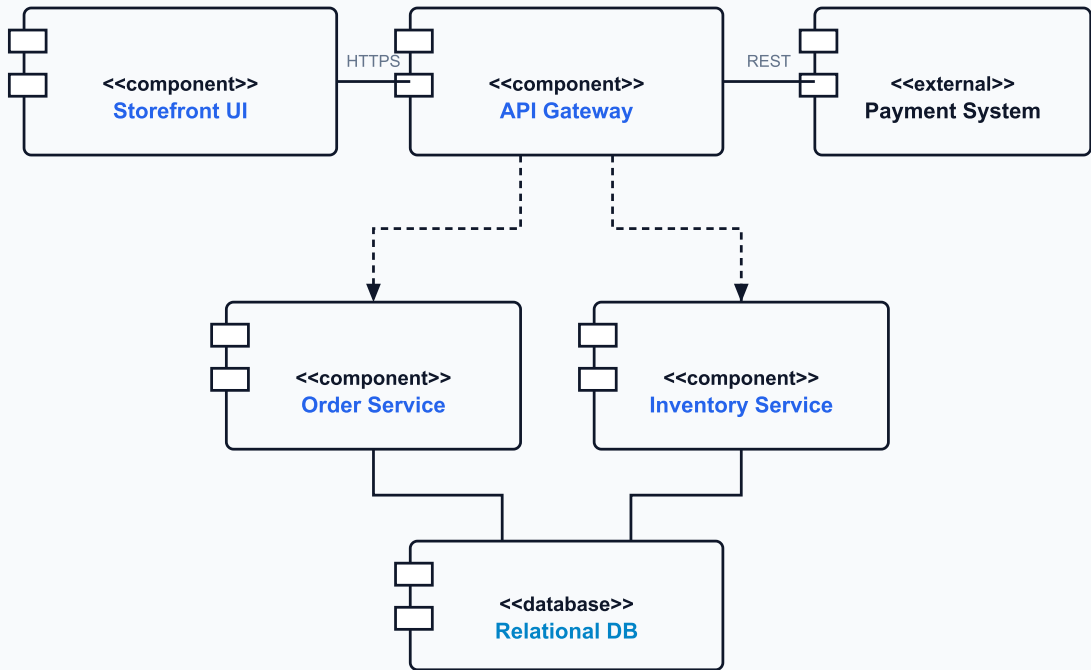


Diagram Key Highlights:

- Storefront UI:** Client-side module communicating exclusively through secure HTTP requests to the Central API Gateway.
- Microservices Layer:** Modular sub-systems (Order Service, Inventory Service) encapsulating core domain business logic.
- External Integration:** Decoupled third-party Payment Service integration via RESTful contracts.

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UML Diagram 04 / 04

4. DEPLOYMENT DIAGRAM

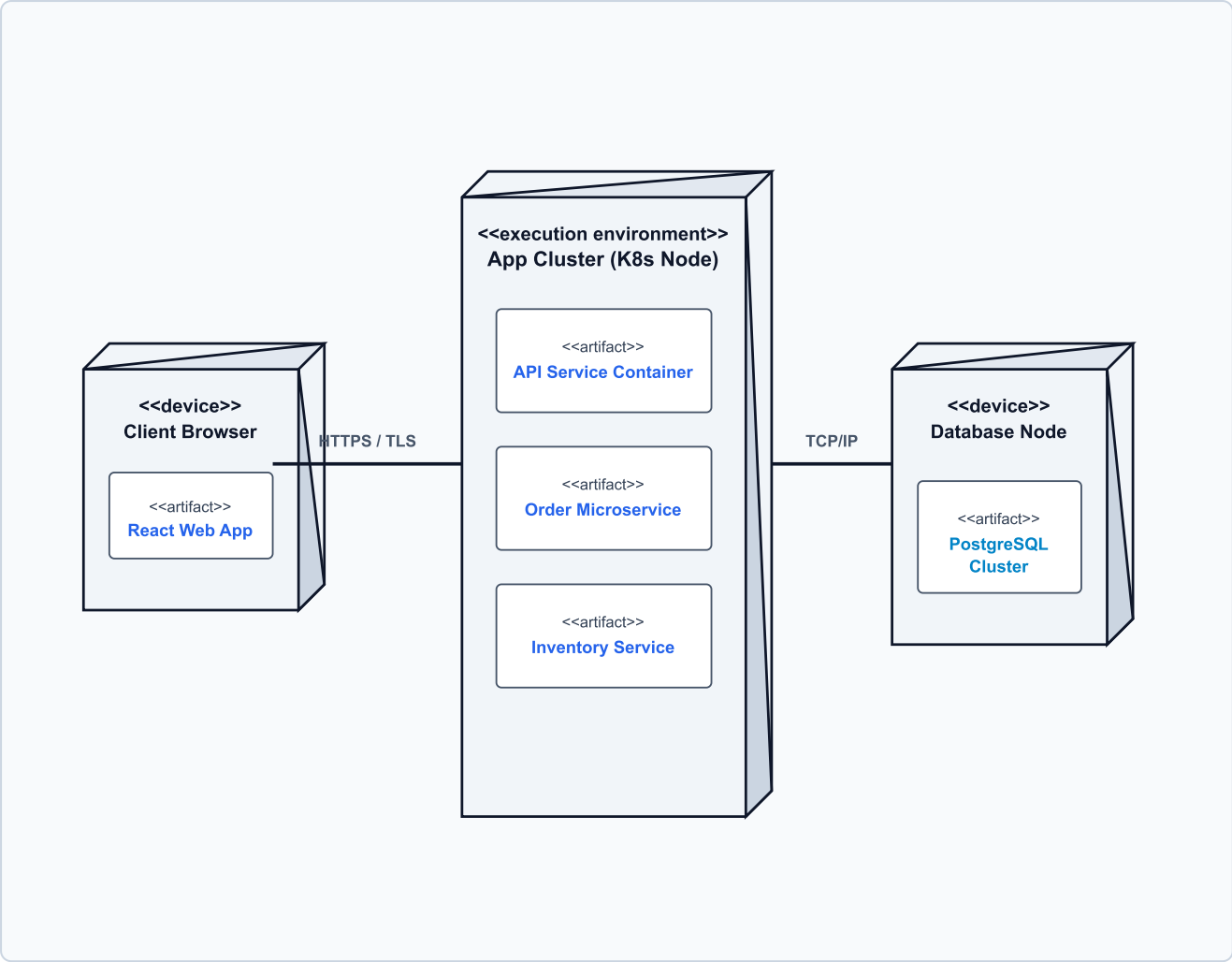


Diagram Key Highlights:

- **Client Tier:** End-user web application executing inside a client web browser node.
- **Application Tier:** Kubernetes-managed cluster hosting containerized microservice artifacts.
- **Data Tier:** Dedicated PostgreSQL database server node communicating with container services via secure internal TCP/IP protocols.